

```
1 "D:\Poojangi\Research Paper\GA for TSP\venv\Scripts\python.exe" "D:/Poojangi/
  Research Paper/GA for TSP/testing2.py"
2 Iteration: 0
3 The best route so far is: [0, 13, 6, 2, 3, 7, 10, 5, 4, 14, 12, 9, 11, 8, 1]
  with distance: 590.0
4 -----
5 Iteration: 1
6 The best route so far is: [0, 13, 6, 2, 3, 7, 10, 5, 4, 14, 12, 9, 11, 8, 1]
  with distance: 590.0
7 -----
8 Iteration: 2
9 The best route so far is: [0, 13, 6, 2, 3, 7, 10, 5, 4, 14, 12, 9, 11, 8, 1]
  with distance: 590.0
10 -----
11 Iteration: 3
12 The best route so far is: [6, 3, 10, 5, 4, 2, 7, 13, 8, 14, 12, 9, 11, 1, 0]
  with distance: 547.0
13 -----
14 Iteration: 4
15 The best route so far is: [6, 3, 10, 5, 4, 2, 7, 13, 8, 14, 12, 9, 11, 1, 0]
  with distance: 547.0
16 -----
17 Iteration: 5
18 The best route so far is: [6, 3, 10, 5, 4, 2, 7, 13, 8, 14, 12, 9, 11, 1, 0]
  with distance: 547.0
19 -----
20 Iteration: 6
21 The best route so far is: [6, 3, 10, 5, 4, 2, 7, 13, 8, 14, 12, 9, 11, 1, 0]
```

```
21 with distance: 547.0
22 -----
23 Iteration: 7
24 The best route so far is: [6, 3, 10, 5, 4, 2, 7, 13, 8, 14, 12, 9, 11, 1, 0]
25 with distance: 547.0
26 -----
27 Iteration: 8
28 The best route so far is: [10, 5, 9, 14, 7, 4, 6, 8, 12, 13, 11, 2, 1, 0, 3]
29 with distance: 535.0
30 -----
31 Iteration: 9
32 The best route so far is: [6, 3, 10, 5, 4, 2, 7, 13, 8, 14, 12, 9, 11, 1, 0]
33 with distance: 547.0
34 -----
35 Iteration: 10
36 The best route so far is: [6, 3, 10, 5, 4, 2, 7, 13, 8, 14, 12, 9, 11, 1, 0]
37 with distance: 547.0
38 -----
39 Iteration: 11
40 The best route so far is: [6, 3, 10, 5, 4, 2, 7, 13, 8, 14, 12, 9, 11, 1, 0]
41 with distance: 547.0
42 -----
43 Iteration: 12
44 The best route so far is: [6, 3, 10, 5, 4, 2, 7, 13, 8, 14, 12, 9, 11, 1, 0]
45 with distance: 547.0
46 -----
47 Iteration: 13
48 The best route so far is: [6, 3, 10, 5, 4, 2, 7, 13, 8, 14, 12, 9, 11, 1, 0]
```

```
42 with distance: 547.0
43 -----
44 Iteration: 14
45 The best route so far is: [5, 2, 13, 14, 7, 4, 6, 8, 12, 9, 11, 1, 0, 3, 10]
   with distance: 547.0
46 -----
47 Iteration: 15
48 The best route so far is: [5, 4, 8, 12, 14, 0, 7, 6, 9, 13, 11, 2, 1, 3, 10]
   with distance: 514.0
49 -----
50 Iteration: 16
51 The best route so far is: [5, 4, 8, 12, 14, 0, 7, 6, 9, 13, 11, 2, 1, 3, 10]
   with distance: 514.0
52 -----
53 Iteration: 17
54 The best route so far is: [5, 4, 8, 12, 14, 0, 7, 6, 9, 13, 11, 2, 1, 3, 10]
   with distance: 514.0
55 -----
56 Iteration: 18
57 The best route so far is: [0, 10, 1, 14, 12, 11, 5, 7, 8, 4, 6, 2, 13, 3, 9]
   with distance: 506.0
58 -----
59 Iteration: 19
60 The best route so far is: [14, 0, 10, 7, 6, 9, 4, 5, 3, 13, 11, 2, 8, 12, 1]
   with distance: 505.0
61 -----
62 Iteration: 20
63 The best route so far is: [14, 0, 10, 7, 6, 9, 4, 5, 3, 13, 11, 2, 8, 12, 1]
```

```
63 with distance: 505.0
64 -----
65 Iteration: 21
66 The best route so far is: [14, 0, 10, 7, 6, 9, 4, 5, 3, 13, 11, 2, 8, 12, 1]
67 with distance: 505.0
68 -----
69 Iteration: 22
69 The best route so far is: [14, 0, 10, 7, 6, 9, 4, 5, 3, 13, 11, 2, 8, 12, 1]
70 with distance: 505.0
71 -----
71 Iteration: 23
72 The best route so far is: [12, 0, 10, 7, 6, 9, 4, 5, 3, 13, 11, 2, 8, 1, 14]
73 with distance: 503.0
74 -----
74 Iteration: 24
75 The best route so far is: [12, 0, 10, 7, 6, 9, 4, 5, 3, 13, 11, 2, 8, 1, 14]
76 with distance: 503.0
77 -----
77 Iteration: 25
78 The best route so far is: [12, 0, 10, 7, 6, 9, 4, 5, 3, 13, 11, 2, 8, 1, 14]
79 with distance: 503.0
80 -----
80 Iteration: 26
81 The best route so far is: [12, 0, 10, 7, 6, 9, 4, 5, 3, 13, 11, 2, 8, 1, 14]
82 with distance: 503.0
83 -----
83 Iteration: 27
84 The best route so far is: [12, 0, 10, 7, 6, 9, 4, 5, 3, 13, 11, 2, 8, 1, 14]
```

```
84 with distance: 503.0
85 -----
86 Iteration: 28
87 The best route so far is: [12, 0, 10, 7, 6, 9, 4, 5, 3, 13, 11, 2, 8, 1, 14]
   with distance: 503.0
88 -----
89 Iteration: 29
90 The best route so far is: [12, 0, 10, 7, 6, 9, 4, 5, 3, 13, 11, 2, 8, 1, 14]
   with distance: 503.0
91 -----
92 Iteration: 30
93 The best route so far is: [0, 10, 8, 7, 5, 11, 9, 3, 13, 2, 6, 4, 1, 14, 12]
   with distance: 476.0
94 -----
95 Iteration: 31
96 The best route so far is: [0, 10, 8, 7, 5, 11, 9, 3, 13, 2, 6, 4, 1, 14, 12]
   with distance: 476.0
97 -----
98 Iteration: 32
99 The best route so far is: [0, 10, 8, 7, 5, 11, 9, 3, 13, 2, 6, 4, 1, 14, 12]
   with distance: 476.0
100 -----
101 Iteration: 33
102 The best route so far is: [0, 10, 8, 7, 5, 11, 9, 3, 13, 2, 6, 4, 1, 14, 12]
   with distance: 476.0
103 -----
104 Iteration: 34
105 The best route so far is: [0, 10, 8, 7, 5, 11, 9, 3, 13, 2, 6, 4, 1, 14, 12]
```

```
105 with distance: 476.0
106 -----
107 Iteration: 35
108 The best route so far is: [0, 10, 8, 7, 5, 11, 9, 3, 13, 2, 6, 4, 1, 14, 12]
109 with distance: 476.0
110 -----
110 Iteration: 36
111 The best route so far is: [0, 10, 8, 7, 5, 11, 9, 3, 13, 2, 6, 4, 1, 14, 12]
112 with distance: 476.0
113 -----
113 Iteration: 37
114 The best route so far is: [0, 10, 8, 7, 5, 11, 9, 3, 13, 2, 6, 4, 1, 14, 12]
115 with distance: 476.0
116 -----
116 Iteration: 38
117 The best route so far is: [0, 10, 8, 13, 3, 5, 7, 11, 9, 2, 6, 4, 1, 14, 12]
118 with distance: 461.0
119 -----
119 Iteration: 39
120 The best route so far is: [0, 10, 8, 13, 3, 5, 7, 11, 9, 2, 6, 4, 1, 14, 12]
121 with distance: 461.0
122 -----
122 Iteration: 40
123 The best route so far is: [8, 13, 3, 5, 7, 11, 9, 2, 6, 4, 1, 14, 12, 0, 10]
124 with distance: 428.0
125 -----
125 Iteration: 41
126 The best route so far is: [8, 13, 3, 5, 7, 11, 9, 2, 6, 4, 1, 14, 12, 0, 10]
```

```
126 with distance: 428.0
127 -----
128 Iteration: 42
129 The best route so far is: [8, 13, 3, 5, 7, 11, 9, 2, 6, 4, 1, 14, 12, 0, 10]
130 with distance: 428.0
131 -----
131 Iteration: 43
132 The best route so far is: [8, 13, 3, 5, 7, 11, 9, 2, 6, 4, 1, 14, 12, 0, 10]
133 with distance: 428.0
134 -----
134 Iteration: 44
135 The best route so far is: [7, 9, 5, 11, 8, 1, 10, 3, 13, 2, 6, 4, 14, 0, 12]
136 with distance: 449.0
137 -----
137 Iteration: 45
138 The best route so far is: [7, 9, 5, 11, 8, 1, 10, 3, 13, 2, 6, 4, 14, 0, 12]
139 with distance: 449.0
140 -----
140 Iteration: 46
141 The best route so far is: [7, 9, 5, 11, 8, 1, 10, 3, 13, 2, 6, 4, 14, 0, 12]
142 with distance: 449.0
143 -----
143 Iteration: 47
144 The best route so far is: [7, 9, 5, 11, 8, 1, 10, 3, 13, 2, 6, 4, 14, 0, 12]
145 with distance: 449.0
146 -----
146 Iteration: 48
147 The best route so far is: [7, 9, 5, 11, 8, 1, 10, 3, 13, 2, 6, 4, 14, 0, 12]
```

```
147 with distance: 449.0
148 -----
149 Iteration: 49
150 The best route so far is: [7, 9, 5, 11, 8, 1, 10, 3, 13, 2, 6, 4, 14, 0, 12]
    with distance: 449.0
151 -----
152 Iteration: 50
153 The best route so far is: [7, 9, 5, 11, 8, 1, 10, 3, 13, 2, 6, 4, 14, 0, 12]
    with distance: 449.0
154 -----
155 Iteration: 51
156 The best route so far is: [7, 9, 5, 11, 8, 1, 10, 3, 13, 2, 6, 4, 14, 0, 12]
    with distance: 449.0
157 -----
158 Iteration: 52
159 The best route so far is: [7, 9, 5, 11, 8, 1, 10, 3, 13, 2, 6, 4, 14, 0, 12]
    with distance: 449.0
160 -----
161 Iteration: 53
162 The best route so far is: [7, 9, 5, 11, 8, 1, 10, 3, 13, 2, 6, 4, 14, 0, 12]
    with distance: 449.0
163 -----
164 Iteration: 54
165 The best route so far is: [7, 9, 5, 11, 8, 1, 10, 3, 13, 2, 6, 4, 14, 0, 12]
    with distance: 449.0
166 -----
167 Iteration: 55
168 The best route so far is: [9, 7, 5, 11, 8, 1, 10, 3, 13, 2, 6, 4, 14, 0, 12]
```



```
168 with distance: 441.0
169 -----
170 Iteration: 56
171 The best route so far is: [9, 7, 5, 11, 8, 1, 10, 3, 13, 2, 6, 4, 14, 0, 12]
172 with distance: 441.0
173 -----
174 Iteration: 57
175 The best route so far is: [9, 7, 5, 11, 8, 1, 10, 3, 13, 2, 6, 4, 14, 0, 12]
176 with distance: 441.0
177 -----
178 Iteration: 58
179 The best route so far is: [9, 7, 5, 11, 8, 1, 10, 3, 13, 2, 6, 4, 14, 0, 12]
180 with distance: 441.0
181 -----
182 Iteration: 60
183 The best route so far is: [9, 7, 5, 11, 8, 1, 10, 3, 13, 2, 6, 4, 14, 0, 12]
184 with distance: 441.0
185 -----
186 Iteration: 61
187 The best route so far is: [9, 7, 5, 11, 8, 6, 2, 13, 3, 10, 1, 4, 14, 0, 12]
188 with distance: 460.0
189 -----
190 Iteration: 62
191 The best route so far is: [9, 7, 5, 11, 8, 6, 2, 13, 3, 10, 1, 4, 14, 0, 12]
```

```
189 with distance: 460.0
190 -----
191 Iteration: 63
192 The best route so far is: [9, 7, 5, 11, 8, 6, 2, 13, 3, 10, 1, 4, 14, 0, 12]
193 with distance: 460.0
194 -----
194 Iteration: 64
195 The best route so far is: [9, 7, 5, 11, 8, 6, 2, 13, 3, 10, 1, 4, 14, 0, 12]
196 with distance: 460.0
197 -----
197 Iteration: 65
198 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 3, 10, 1, 4, 14, 0, 12]
199 with distance: 444.0
200 -----
200 Iteration: 66
201 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 3, 10, 1, 4, 14, 0, 12]
202 with distance: 444.0
203 -----
203 Iteration: 67
204 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 3, 10, 1, 4, 14, 0, 12]
205 with distance: 444.0
206 -----
206 Iteration: 68
207 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 3, 10, 1, 4, 14, 0, 12]
208 with distance: 444.0
209 -----
209 Iteration: 69
210 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 3, 10, 1, 4, 14, 0, 12]
```

```
210 with distance: 444.0
211 -----
212 Iteration: 70
213 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 3, 10, 1, 4, 14, 0, 12]
214 with distance: 444.0
215 -----
216 Iteration: 71
217 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 3, 10, 1, 4, 14, 0, 12]
218 with distance: 444.0
219 -----
220 Iteration: 72
221 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 3, 10, 1, 4, 14, 0, 12]
222 with distance: 444.0
223 -----
224 Iteration: 74
225 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 3, 10, 1, 4, 14, 0, 12]
226 with distance: 444.0
227 -----
228 Iteration: 75
229 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 3, 10, 1, 4, 14, 0, 12]
230 with distance: 444.0
231 -----
232 Iteration: 76
233 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 3, 10, 1, 4, 14, 0, 12]
```

```
231 with distance: 444.0
232 -----
233 Iteration: 77
234 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 3, 10, 1, 4, 14, 0, 12]
    with distance: 444.0
235 -----
236 Iteration: 78
237 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 3, 10, 1, 4, 14, 0, 12]
    with distance: 444.0
238 -----
239 Iteration: 79
240 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 3, 10, 1, 4, 14, 0, 12]
    with distance: 444.0
241 -----
242 Iteration: 80
243 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 3, 10, 1, 4, 14, 0, 12]
    with distance: 444.0
244 -----
245 Iteration: 81
246 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 3, 10, 1, 4, 14, 0, 12]
    with distance: 444.0
247 -----
248 Iteration: 82
249 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 3, 10, 1, 4, 14, 0, 12]
    with distance: 444.0
250 -----
251 Iteration: 83
252 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 3, 10, 1, 4, 14, 0, 12]
```

```
252 with distance: 444.0
253 -----
254 Iteration: 84
255 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 3, 10, 1, 4, 14, 0, 12]
    with distance: 444.0
256 -----
257 Iteration: 85
258 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 3, 10, 1, 4, 14, 0, 12]
    with distance: 444.0
259 -----
260 Iteration: 86
261 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 3, 10, 1, 4, 14, 0, 12]
    with distance: 444.0
262 -----
263 Iteration: 87
264 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 3, 10, 1, 4, 14, 0, 12]
    with distance: 444.0
265 -----
266 Iteration: 88
267 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 3, 10, 1, 4, 14, 0, 12]
    with distance: 444.0
268 -----
269 Iteration: 89
270 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 3, 10, 1, 4, 8, 14, 0, 12]
    with distance: 440.0
271 -----
272 Iteration: 90
273 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 3, 10, 1, 4, 8, 14, 0, 12]
```

```
273 with distance: 440.0
274 -----
275 Iteration: 91
276 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 3, 10, 1, 4, 14, 0, 12]
277 with distance: 444.0
278 -----
279 Iteration: 92
280 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 3, 10, 1, 4, 14, 0, 12]
281 with distance: 444.0
282 -----
283 Iteration: 93
284 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 3, 10, 1, 4, 14, 0, 12]
285 with distance: 444.0
286 -----
287 Iteration: 94
288 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 3, 10, 1, 4, 14, 0, 12]
289 with distance: 444.0
290 -----
291 Iteration: 95
292 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 3, 10, 1, 4, 14, 0, 12]
293 with distance: 444.0
294 -----
295 Iteration: 96
296 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 3, 10, 1, 4, 14, 0, 12]
297 with distance: 444.0
298 -----
299 Iteration: 97
300 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 3, 10, 1, 4, 14, 0, 12]
```

```
294 with distance: 444.0
295 -----
296 Iteration: 98
297 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 3, 10, 1, 4, 14, 0, 12]
298 with distance: 444.0
299 -----
299 Iteration: 99
300 The best route so far is: [9, 7, 8, 6, 2, 13, 11, 5, 3, 10, 1, 4, 14, 0, 12]
301 with distance: 407.0
302 -----
302 Iteration: 100
303 The best route so far is: [9, 7, 8, 6, 2, 13, 11, 5, 3, 10, 1, 4, 14, 0, 12]
304 with distance: 407.0
305 -----
305 Iteration: 101
306 The best route so far is: [9, 7, 8, 6, 2, 13, 11, 5, 3, 10, 1, 4, 14, 0, 12]
307 with distance: 407.0
308 -----
308 Iteration: 102
309 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 3, 10, 1, 4, 14, 0, 12]
310 with distance: 444.0
311 -----
311 Iteration: 103
312 The best route so far is: [5, 12, 1, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 9, 7]
313 with distance: 440.0
314 -----
314 Iteration: 104
315 The best route so far is: [5, 12, 1, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 9, 7]
```

```
315 with distance: 440.0
316 -----
317 Iteration: 105
318 The best route so far is: [12, 1, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 9, 7]
    with distance: 418.0
319 -----
320 Iteration: 106
321 The best route so far is: [12, 1, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 9, 7]
    with distance: 418.0
322 -----
323 Iteration: 107
324 The best route so far is: [12, 1, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 9, 7]
    with distance: 418.0
325 -----
326 Iteration: 108
327 The best route so far is: [12, 1, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 9, 7]
    with distance: 418.0
328 -----
329 Iteration: 109
330 The best route so far is: [12, 1, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 9, 7]
    with distance: 418.0
331 -----
332 Iteration: 110
333 The best route so far is: [12, 1, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 9, 7]
    with distance: 418.0
334 -----
335 Iteration: 111
336 The best route so far is: [12, 1, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 9, 7]
```



```
336 with distance: 418.0
337 -----
338 Iteration: 112
339 The best route so far is: [12, 1, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 9, 7]
340 with distance: 418.0
341 -----
341 Iteration: 113
342 The best route so far is: [12, 1, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 9, 7]
343 with distance: 418.0
344 -----
344 Iteration: 114
345 The best route so far is: [1, 9, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 7, 12]
346 with distance: 414.0
347 -----
347 Iteration: 115
348 The best route so far is: [1, 9, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 7, 12]
349 with distance: 414.0
350 -----
350 Iteration: 116
351 The best route so far is: [1, 9, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 7, 12]
352 with distance: 414.0
353 -----
353 Iteration: 117
354 The best route so far is: [1, 9, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 7, 12]
355 with distance: 414.0
356 -----
356 Iteration: 118
357 The best route so far is: [5, 11, 13, 9, 7, 2, 6, 8, 4, 14, 0, 10, 3, 12, 1]
```

```
357 with distance: 406.0
358 -----
359 Iteration: 119
360 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 1, 12]
361 with distance: 383.0
362 -----
363 Iteration: 120
364 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 1, 12]
365 with distance: 383.0
366 -----
367 Iteration: 121
368 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 1, 12]
369 with distance: 383.0
370 -----
371 Iteration: 123
372 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 1, 12]
373 with distance: 383.0
374 -----
375 Iteration: 124
376 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 1, 12]
377 with distance: 383.0
378 -----
379 Iteration: 125
380 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 1, 12]
```

```
378 with distance: 383.0
379 -----
380 Iteration: 126
381 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 1, 12]
382 with distance: 383.0
383 -----
383 Iteration: 127
384 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 1, 12]
385 with distance: 383.0
386 -----
386 Iteration: 128
387 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 1, 12]
388 with distance: 383.0
389 -----
389 Iteration: 129
390 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 1, 12]
391 with distance: 383.0
392 -----
392 Iteration: 130
393 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 1, 12]
394 with distance: 383.0
395 -----
395 Iteration: 131
396 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 1, 12]
397 with distance: 383.0
398 -----
398 Iteration: 132
399 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 1, 12]
```

```
399 with distance: 383.0
400 -----
401 Iteration: 133
402 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 1, 12]
    with distance: 383.0
403 -----
404 Iteration: 134
405 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 1, 12]
    with distance: 383.0
406 -----
407 Iteration: 135
408 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 1, 12]
    with distance: 383.0
409 -----
410 Iteration: 136
411 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 1, 12]
    with distance: 383.0
412 -----
413 Iteration: 137
414 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 1, 12]
    with distance: 383.0
415 -----
416 Iteration: 138
417 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 1, 12]
    with distance: 383.0
418 -----
419 Iteration: 139
420 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 1, 12]
```

```
420 with distance: 383.0
421 -----
422 Iteration: 140
423 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 1, 12]
424 with distance: 383.0
425 -----
426 Iteration: 141
427 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 1, 12]
428 with distance: 383.0
429 -----
430 Iteration: 142
431 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 1, 12]
432 with distance: 383.0
433 -----
434 Iteration: 143
435 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 1, 12]
436 with distance: 383.0
437 -----
438 Iteration: 144
439 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 1, 12]
440 with distance: 383.0
441 -----
442 Iteration: 145
443 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 1, 12]
444 with distance: 383.0
445 -----
446 Iteration: 146
447 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 1, 12]
```

```
441 with distance: 383.0
442 -----
443 Iteration: 147
444 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 1, 12]
445 with distance: 383.0
446 -----
447 Iteration: 148
448 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 1, 12]
449 with distance: 383.0
450 -----
451 Iteration: 149
452 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 1, 12]
453 with distance: 383.0
454 -----
455 Iteration: 150
456 The best route so far is: [9, 7, 5, 11, 13, 2, 6, 8, 4, 14, 0, 10, 3, 1, 12]
457 with distance: 383.0
458 -----
459 Iteration: 151
460 The best route so far is: [12, 0, 14, 11, 13, 2, 6, 8, 4, 1, 9, 7, 5, 3, 10]
461 with distance: 409.0
462 -----
463 Iteration: 152
464 The best route so far is: [14, 7, 5, 11, 13, 2, 6, 8, 4, 1, 9, 3, 10, 12, 0]
465 with distance: 407.0
466 -----
467 Iteration: 153
468 The best route so far is: [14, 7, 5, 11, 13, 2, 6, 8, 4, 1, 9, 3, 10, 12, 0]
```

```
462 with distance: 407.0
463 -----
464 Iteration: 154
465 The best route so far is: [14, 7, 5, 11, 13, 2, 6, 8, 4, 1, 9, 3, 10, 12, 0]
466 with distance: 407.0
467 -----
468 Iteration: 155
469 The best route so far is: [12, 14, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 1, 0]
470 with distance: 359.0
471 -----
472 Iteration: 156
473 The best route so far is: [0, 12, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 1]
474 with distance: 342.0
475 -----
476 Iteration: 157
477 The best route so far is: [0, 12, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 1]
478 with distance: 342.0
479 -----
480 Iteration: 158
481 The best route so far is: [0, 12, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 1]
482 with distance: 342.0
483 -----
484 Iteration: 159
485 The best route so far is: [12, 0, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 1]
486 with distance: 329.0
487 -----
488 Iteration: 160
489 The best route so far is: [12, 0, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 1]
```

```
483 with distance: 329.0
484 -----
485 Iteration: 161
486 The best route so far is: [12, 0, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 1]
487 with distance: 329.0
488 -----
488 Iteration: 162
489 The best route so far is: [12, 0, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 1]
490 with distance: 329.0
491 -----
491 Iteration: 163
492 The best route so far is: [12, 0, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 1]
493 with distance: 329.0
494 -----
494 Iteration: 164
495 The best route so far is: [12, 0, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 1]
496 with distance: 329.0
497 -----
497 Iteration: 165
498 The best route so far is: [12, 0, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 1]
499 with distance: 329.0
500 -----
500 Iteration: 166
501 The best route so far is: [12, 0, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 1]
502 with distance: 329.0
503 -----
503 Iteration: 167
504 The best route so far is: [12, 0, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 1]
```



```
504 with distance: 329.0
505 -----
506 Iteration: 168
507 The best route so far is: [12, 0, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 1]
508 with distance: 329.0
509 -----
509 Iteration: 169
510 The best route so far is: [12, 0, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 1]
511 with distance: 329.0
512 -----
512 Iteration: 170
513 The best route so far is: [12, 0, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 1]
514 with distance: 329.0
515 -----
515 Iteration: 171
516 The best route so far is: [12, 0, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 1]
517 with distance: 329.0
518 -----
518 Iteration: 172
519 The best route so far is: [12, 0, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 1]
520 with distance: 329.0
521 -----
521 Iteration: 173
522 The best route so far is: [12, 0, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 1]
523 with distance: 329.0
524 -----
524 Iteration: 174
525 The best route so far is: [12, 0, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 1]
```

```
525 with distance: 329.0
526 -----
527 Iteration: 175
528 The best route so far is: [12, 0, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 1]
529 with distance: 329.0
529 -----
530 Iteration: 176
531 The best route so far is: [12, 0, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 1]
531 with distance: 329.0
532 -----
533 Iteration: 177
534 The best route so far is: [12, 0, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 1]
534 with distance: 329.0
535 -----
536 Iteration: 178
537 The best route so far is: [12, 0, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 1]
537 with distance: 329.0
538 -----
539 Iteration: 179
540 The best route so far is: [12, 0, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 1]
540 with distance: 329.0
541 -----
542 Iteration: 180
543 The best route so far is: [12, 1, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 0]
543 with distance: 325.0
544 -----
545 Iteration: 181
546 The best route so far is: [12, 1, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 0]
```

```
546 with distance: 325.0
547 -----
548 Iteration: 182
549 The best route so far is: [12, 1, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 0]
550 with distance: 325.0
551 -----
552 Iteration: 183
553 The best route so far is: [12, 1, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 0]
554 with distance: 325.0
555 -----
556 Iteration: 184
557 The best route so far is: [12, 1, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 0]
558 with distance: 325.0
559 -----
560 Iteration: 185
561 The best route so far is: [12, 1, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 0]
562 with distance: 325.0
563 -----
564 Iteration: 186
565 The best route so far is: [12, 1, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 0]
566 with distance: 325.0
567 -----
568 Iteration: 187
569 The best route so far is: [12, 1, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 0]
570 with distance: 325.0
571 -----
572 Iteration: 188
573 The best route so far is: [12, 1, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 0]
```

```
567 with distance: 325.0
568 -----
569 Iteration: 189
570 The best route so far is: [12, 1, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 0]
571 with distance: 325.0
572 -----
573 Iteration: 190
574 The best route so far is: [12, 1, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 0]
575 with distance: 325.0
576 -----
577 Iteration: 191
578 The best route so far is: [12, 1, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 0]
579 with distance: 325.0
580 -----
581 Iteration: 193
582 The best route so far is: [12, 1, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 0]
583 with distance: 325.0
584 -----
585 Iteration: 194
586 The best route so far is: [12, 1, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 0]
587 with distance: 325.0
588 -----
589 Iteration: 195
590 The best route so far is: [12, 1, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 0]
```

```
588 with distance: 325.0
589 -----
590 Iteration: 196
591 The best route so far is: [12, 1, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 0]
592 with distance: 325.0
593 -----
594 Iteration: 197
595 The best route so far is: [12, 1, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 0]
596 with distance: 325.0
597 -----
598 Iteration: 198
599 The best route so far is: [12, 1, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 0]
600 with distance: 325.0
601 -----
602 Iteration: 199
603 The best route so far is: [12, 1, 10, 3, 5, 7, 9, 11, 13, 2, 6, 8, 4, 14, 0]
604 with distance: 325.0
605 -----
606 End
607 -----
608 Process finished with exit code 0
```